



Introduction to Linux and High Performance Computing

Gloria Nakabiri, David Twesigomwe, Ian Wasukira

What is Linux?

Linux is a free open-source derivative of the Unix Operating System (OS).

Unix

Original family of operating systems—often proprietary and commercially licensed

Distributions: IBM AIX, HP-UX, Sun Solaris



Linux

Open source OS based on Unix. It runs on computers, servers, mobile phones and a wide range of other devices

Distributions: Ubuntu, Redhat, Fedora, Debian, Android

Importance of Linux

- Offers high security and stability, making it ideal for servers and development work – supporting safe data storage & **multi-user systems**
- Fully open-source and free to use, modify, and distribute.
- Highly flexible and customisable to suit different user and industry needs – **portability**
- Supported by a large global community and a vast software ecosystem

Where is Linux used?



**Servers and
Hosting**



Development



**Desktop And
Personal Use**



Cybersecurity

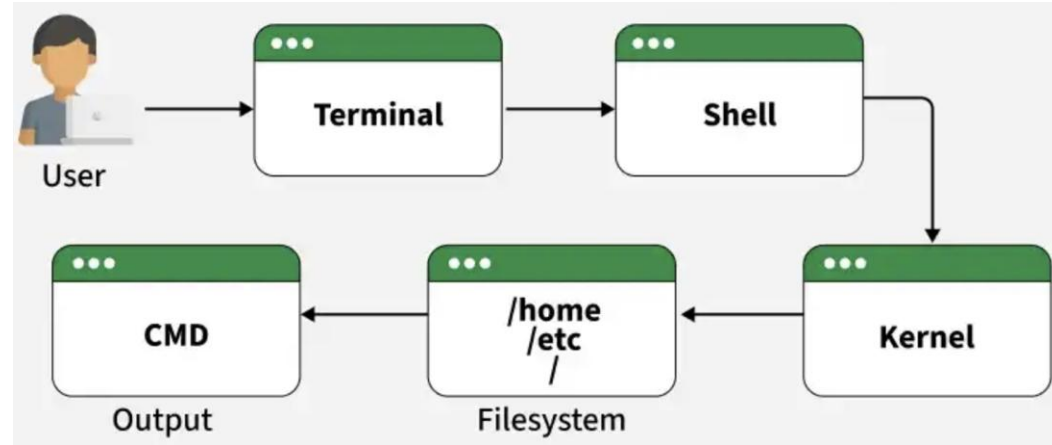
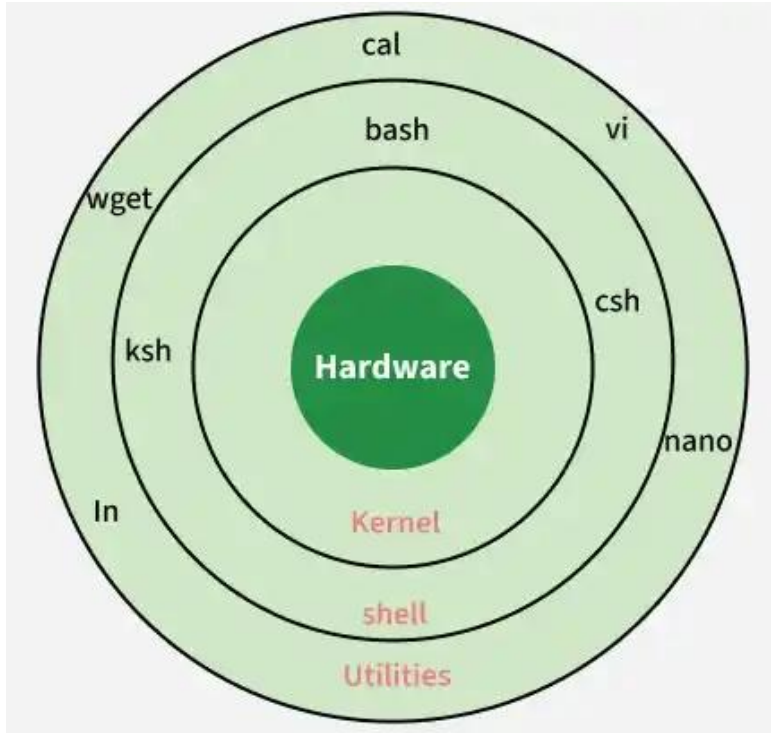


**Embedded
Systems**

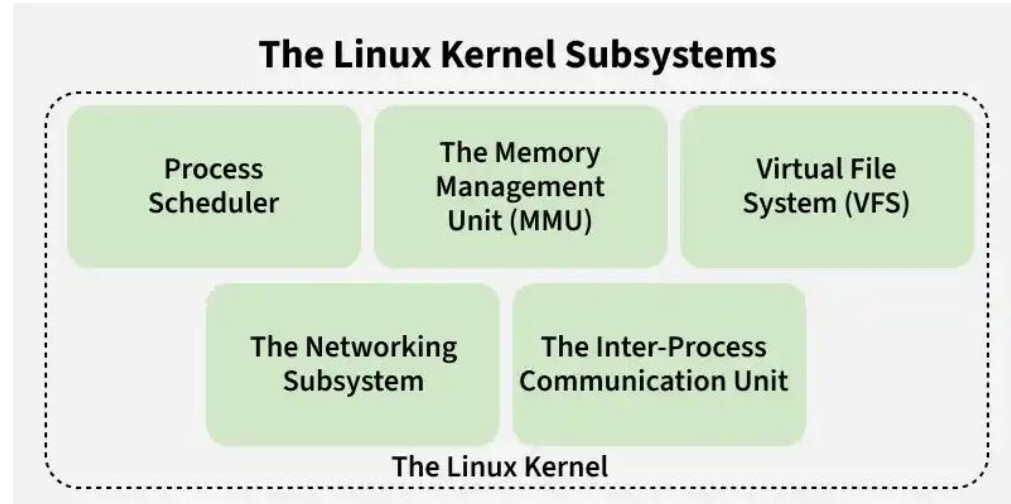
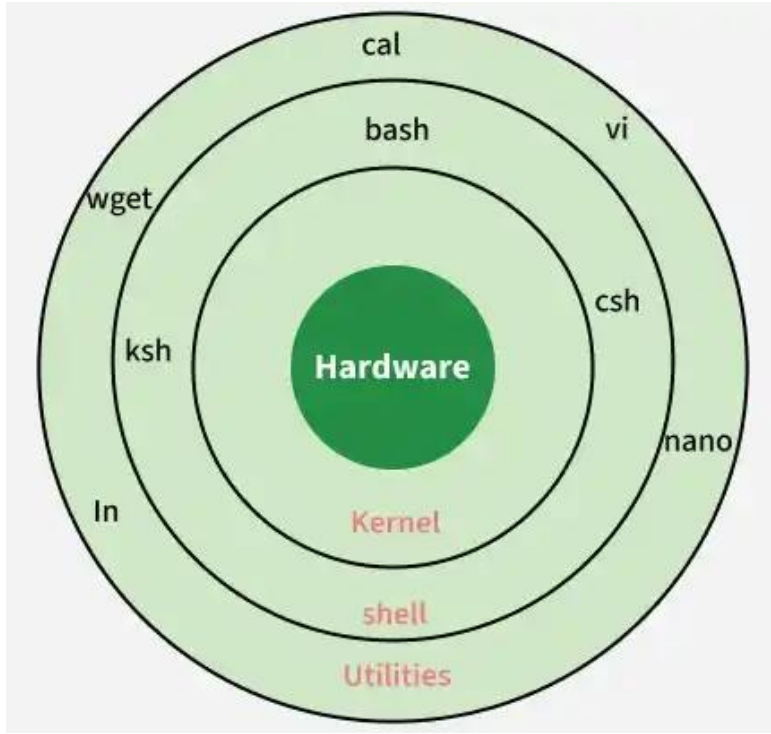


Supercomputers

Basic Linux Architecture

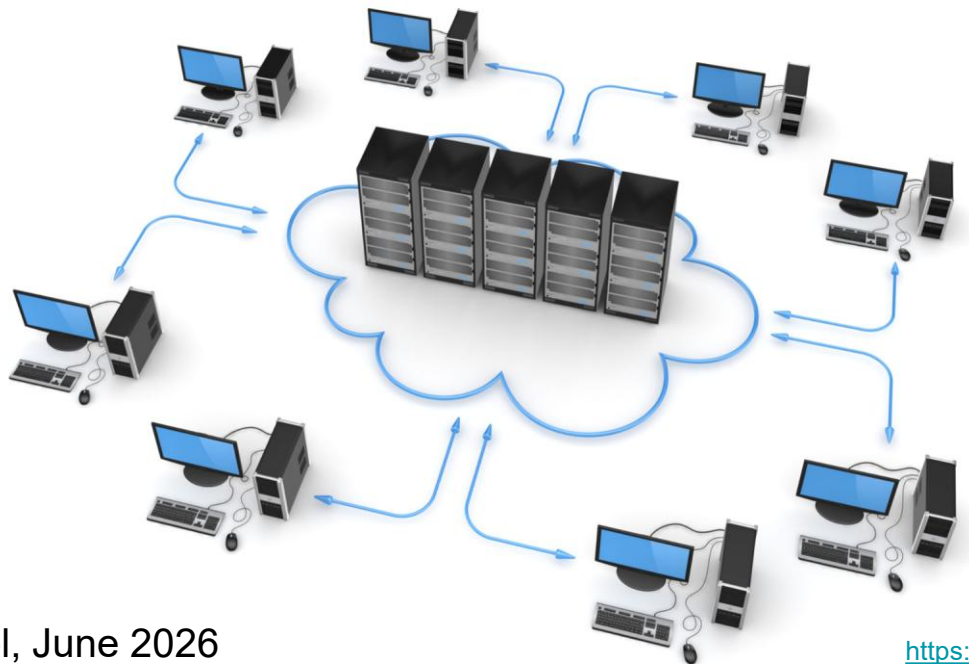


Basic Linux Architecture



What is high-performance computing (HPC)?

- Technology that uses clusters of powerful processors that work in parallel to process massive, multidimensional data sets and solve complex problems at extremely high speeds.



Importance of high performance computing

- Speed: Reduces processing times for large genomic datasets (through parallelisation and fast networks across elements)
- Scalability: Scales to meet the growing demand for data storage and processing.
- Accuracy: Improves the reliability of results through efficient data standardisation

Use cases of high performance computing

- Large Scale Genome Assembly and Read Alignment
- Large-scale Variant Calling and Annotation e.g. in population genetics and pathogen surveillance
- Pathogen genomics and Microbiome studies
- Protein Structure Prediction and Molecular Dynamics
- Machine Learning Integration to detect complex patterns within omic datasets

Challenges in high performance computing

- **Cost:** The cost of the hardware, software, and energy consumption is enormous, making HPC systems exceedingly expensive to create and operate. Additionally, the setup and management of HPC systems require qualified workers, which raises the overall cost.
- **Scalability:** HPC systems must be made scalable so they may be modified or expanded as necessary to meet shifting demands.
- **Data Management:** Large volumes of data must be stored securely and accessed using sophisticated networking analysis infrastructure

Challenges in high performance computing

- **Programming:** It might be challenging for developers to learn how to create and optimise algorithms for parallel processing.
- **Support for software and tools:** HPC systems need specific software and tools to function efficiently — not all software and tools are created to function with HPC equipment.
- **Power consumption and cooling:** To maintain the hardware functioning at its best, specialised cooling technologies are needed for HPC systems' high heat production. Furthermore, HPC systems consume a lot of electricity

Practical 1

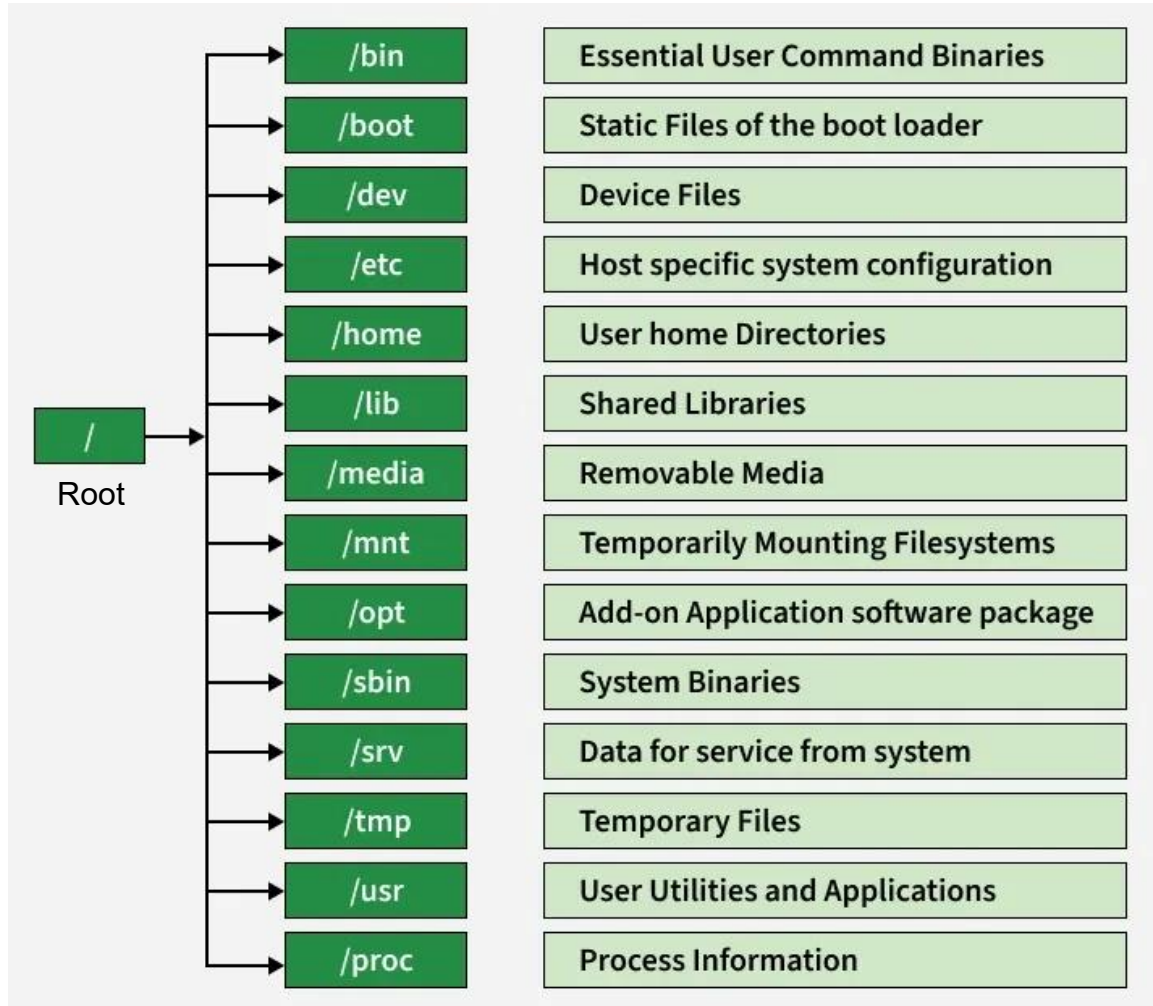
Generating ssh keys

Logging into the ACE-Uganda HPC cluster

Basic cluster architecture

- Login node(s) vs worker nodes
- Directory structure on shared systems, shared storage

Linux / HPC file structure



Practical 1: continued

Task 1: Explore the ACE cluster filesystem, check available modules (module avail), load one (conda), and observe how the environment changes.